Lauren Harrison

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EDUCATION

B.Sc. Honours in Physics and Astronomy | Minor in Mathematics Sept 2022 – Present
University of Victoria Victoria, BC
Supervisor: Prof. Kim Venn

- **Dean's List**, 2025–2026
- Relevant Coursework: Stellar Astrophysics, Observational Astronomy, Astrophysics, Computer Assisted Physics, Computational Modeling and Analysis, Planetary Science

PUBLICATIONS

1. McConnachie et al. (including **Harrison, L.**). The GHOST Ultra-faint dwarf Legacy Survey (GHOULS) I. Survey Design and First Results (2026, In prep.)
2. Dovgal & Venn et al. (including **Harrison, L.**). The discovery of the most pristine star found in its birth environment, the ultrafaint dwarf galaxy Carina II. *Nature* (2026, Submitted)

RESEARCH & WORK EXPERIENCE

Astronomy Research Assistant (Co-op) September 2026 – Present
NRC Herzberg Astronomy & Astrophysics Victoria, BC
Supervisor: Dr. Deborah Lokhorst

- Analyzing wide-field polarimetry data of the M82 galaxy from the Dragonfly Telephoto Array to characterize faint and extended galactic structures.

Trottier Summer Research Intern (Co-op) May 2026 – August 2026
Trottier Institute for Research on Exoplanets, McGill University Montréal, QC
Supervisor: Prof. Nicolas Cowan

- Ran spectral sampling jobs on GPUs using Digital Research Alliance of Canada computing resources.
- Updated a PyTorch pipeline from synthetic to real observations of Barnard's Star using SPIRou observations.
- Used trained diffusion machine learning models to perform sampling of stellar spectra to further constrain the masses of exoplanets.

Junior Astronomy Research Assistant May 2025 – August 2025
University of Victoria Victoria, BC
Supervisor: Prof. Kim Venn

- Reduced large spectroscopic datasets from Gemini South, integrating a radial velocity correction in Python into stellar spectra for $50+$ stars included in GHOULS (GHOST Ultrafaint Legacy Survey).
- Implemented a Python-based algorithm to analyze stellar spectra by measuring equivalent widths from Fe I/II line lists, infer stellar parameters with uncertainties, and catalog results.
- Worked in a Linux/Unix environment on remote servers, using SSH and command-line tools for data transfer and execution of spectroscopic reduction and analysis pipelines.

Directed Studies Research Course May 2025 – June 2025
University of Victoria Victoria, BC
Supervisor: Dr. Stephenson Yang

- Reduced and cleaned 98 spectral observations in IRAF of the variable and binary star, W Cephei. Processing included cosmic ray removal, continuum fitting, and wavelength calibration.
- Implemented a Python-based pipeline to derive precise radial velocities and orbital parameters, using Lomb–Scargle periodograms. Improved precision through automated telluric line removal.
- Investigated a potential 776-day periodicity through comparative time-series analysis of spectroscopic and photometric datasets.

Astronomy Open House Facilitator Sept 2024 – August 2025
University of Victoria Victoria, BC
Supervisor: Dr. Erica Franzmann

- Facilitated over 35 public and private tours of the University of Victoria's observatory and facilitated night observing during the observatory's weekly open house. Engaged with a diverse audience of up to 50 weekly attendees.
- Organized public outreach programs, including delivering night sky tours and educational sessions to various audiences. Maintained social media accounts (Instagram, Facebook) and increased followers by $6\,\%$.

Video Developer in Astronomical Concepts (Co-op) May 2024 – August 2024
NRC Herzberg Astronomy & Astrophysics Victoria, BC
Supervisors: Dr. Brenda Matthews & Dr. Gerald Schieven

- Produced short (5-15 minute) educational videos tailored for astronomers new to radio astronomy using the Adobe Suite software, including creating simulated spectra in Adobe Illustrator.
- Created 4 high-quality narrated videos for YouTube and the National Radio Astronomy Observatory (NRAO) ALMA Science Portal, ensuring concepts were communicated to a global audience.
- Collaborated with researchers to storyboard and fact-check content. Ensured scientific accuracy while meeting the outreach objectives of the National Radio Astronomy Observatory.

TECHNICAL SKILLS

- **Programming Languages:** Python, Julia, HTML, R, SQL
- **Computing Environments:** Linux/Unix (command line, bash), Windows
- **Astronomical Tools:** IRAF, SAOImageDS9, FITS file processing
- **Software:** GitHub, Jupyter Notebook, Microsoft Office Suite

COMMUNITY ENGAGEMENT & OUTREACH

Student Mentor Sept 2026 - Present
First Year Interconnect Program, University of Victoria Victoria, BC

- Mentor 3 first-year engineering students and provide academic, personal, and university-transition support throughout their first year. Provide relevant resources based on their individuals interests.

Invited Outreach Facilitator Sept 2026
Centre of the Universe / Nimmo Bay Resort Nimmo Bay, BC

- Delivered a 2-night remote astronomy and stargazing outreach program in a wilderness setting with near-zero light pollution. Conducted guided night-sky tours to 13 guests.

Outreach Volunteer May 2026 – August 2026
Trottier Institute for Research on Exoplanets, McGill University Montréal, QC

- Assisted with public astronomy outreach events, engaging attendees and explaining exoplanet research concepts to a diverse audience.

EDI Physics & Astronomy Undergraduate Representative April 2026 – Present
University of Victoria Victoria, BC

- Collaborated with committee members to advance equity, diversity, and inclusion initiatives within the department.

Vice President of Marketing Sept 2025 – Present
Physics and Astronomy Students Society, University of Victoria Victoria, BC

- Collaborate with a 6-person executive team of the Physics and Astronomy Course Union to coordinate student engagement and events, with a focus on involving 1st- and 2nd-year students.
- Facilitate and promote society events, meetings, and fundraisers to the Physics and Astronomy student community on various social media platforms.

Volunteer Tour Guide Sept 2024 – Present
Centre of the Universe, Dominion Astrophysical Observatory Victoria, BC

- Deliver informative and interactive tours of the 1.83m Plaskett Telescope to public and private events, explaining astronomical concepts to audiences with varying levels of expertise.
- Assisted with over 30 Star Parties and public outreach events hosting 200+ attendees, answering questions regarding the history and science of the site.
- Selected as a lead volunteer to provide specialized observatory tours for international cruise ship guests to enhance global outreach of the site.

PRESENTATIONS

TSI Summer Research Showcase August 2026
Trottier Space Institute, McGill University Montréal, QC
Awarded Session's Best Presenter Award

iREx Summer Research Showcase August 2026
Trottier Institute for Research on Exoplanets, Université de Montréal Montréal, QC

International Astronomy Day: Women Who Reached for the Stars April 2026
Royal BC Museum Victoria, BC

Centre of the Universe Star Party: Women Who Reached for the Stars March 2026
Dominion Astrophysical Observatory Victoria, BC

Centre of the Universe Star Party: GHOSTS and GHOULS December 2025
Dominion Astrophysical Observatory Victoria, BC

NRC Herzberg Astronomy and Astrophysics Science Tea August 2024
Dominion Astrophysical Observatory Victoria, BC